

# Long Ma

## Postdoctoral Research Fellow

CSE, CUHK

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🌐 Homepage

📍 G/F, Pentecostal Mission Hall

### Profile

- During my PhD, I co-supervised a master's paper that received the PRCV 2022 Best Poster Award, and my doctoral dissertation received the **CSIG Doctoral Dissertation Incentive Program Nomination Award** (with only 20 theses selected nationwide).
- According to Google Scholar, my work has received over **6,000 citations** with an h-index of 30, including two CVPR papers (one as first author and one as second author) that have each been cited more than 1,400 times.

### Research Interests

My research focuses on visual perception methods and the design of learning strategies for adverse conditions. Currently, my work centers on developing novel learning paradigms for vision tasks based on bilevel optimization.

### Education

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|---------------------------------------------------------------------------------------------------|------------------------------|-----------------|
| <b>Dalian University of Technology</b> <i>Postdoctoral Research Fellow</i>                        | China                        | 05/2023–05/2026 |
| Thesis: Bilevel Optimization for Vision Perception under Adverse Conditions                       | Co-Supervisor: Zhongxuan Luo |                 |
| <b>Dalian University of Technology</b> <i>PhD in Software Engineering</i>                         | China                        | 09/2019–03/2023 |
| Thesis: Research on Low-Light Vision Perception                                                   | Advisor: Risheng Liu         |                 |
| <b>Dalian University of Technology</b> <i>Master in Software Engineering</i>                      | China                        | 09/2016–06/2019 |
| Thesis: Optimization-driven Deep Learning for Image Enhancement                                   | Advisor: Risheng Liu         |                 |
| <b>Northeast Agricultural University</b> <i>Bachelor in Information and Computational Science</i> | China                        | 09/2012–06/2016 |
| Thesis: Image Restoration based on Generalized Shrinkage Operators                                | Advisor: Qiufeng Wu          |                 |

### Selected Publications (†Corresponding Author)

- **Long Ma**, Tengyu Ma, Chengpei Xu, Jinyuan Liu, Xin Fan, Zhongxuan Luo, Risheng Liu. *Learning with Self-Calibrator for Fast and Robust Low-Light Image Enhancement*. **IEEE TPAMI** 2025, 47(10): 9095-9112. (JCR-Q1, IF: 20.4)
- **Long Ma**, Dian Jin, Nan An, Jinyuan Liu, Xin Fan, Zhongxuan Luo, Risheng Liu. *Bilevel Fast Scene Adaptation for Low-Light Image Enhancement*. **IJCV** 2025, 133: 8234-8252. (JCR-Q1, IF: 10.3)
- **Long Ma**, Nan An, Jinyuan Liu, Xin Fan, Zhongxuan Luo, Deyu Meng, Risheng Liu. *Practical Exposure Correction via Compensation*. **Science China Information Science** 2025. (Accepted) (JCR-Q1, IF: 8.1)
- **Long Ma**, Haoze Zheng, Yuhang Mao, Jinyuan Liu, Chengpei Xu, Xinwei Xue, Yi Wang, Xiangjian He, Weimin Wang. *BiPA: Bilevel Prompt Adaptation for Underwater Instance Segmentation*. **CVPR** 2026.
- **Long Ma**, Yuxin Feng, Yan Zhang, Jinyuan Liu, Weimin Wang, Guang-Yong Chen, Chengpei Xu, Zhuo Su. *CoA: Towards Real Image Dehazing via Compression-and-Adaptation*. **CVPR** 2025: 11197-11206.
- **Long Ma**, Tengyu Ma, Risheng Liu, Xin Fan, Zhongxuan Luo. *Toward Fast, Flexible, and Robust Low-Light Image Enhancement*. **CVPR** 2022: 5637-5646. (Oral)
- **Long Ma**, Risheng Liu, Jiaao Zhang, Xin Fan, Zhongxuan Luo. *Learning Deep Context-Sensitive Decomposition for Low-Light Image Enhancement*. **IEEE TNNLS** 2021, 33(10): 5666-5680. (JCR-Q1, IF: 9.7)
- **Long Ma**, Risheng Liu, Yiyang Wang, Xin Fan, Zhongxuan Luo. *Low-Light Image Enhancement via Self-Reinforced Retinex Projection Model*. **IEEE TMM** 2022, 25: 3573-3586. (JCR-Q1, IF: 9.9)
- Jialun Pei, Diandian Guo, Donghui Yang, Zhixi Li, Yuxin Feng, **Long Ma**<sup>†</sup>, Bo Du, Pheng-Ann Heng. *Benchmarking Endoscopic Surgical Image Restoration and Beyond*. **CVPR** 2026.
- Xinwei Xue, Zimeng Xu, Kaixin Wang, Jincheng Yuan, Jingchun Zhou, Chengpei Xu, Xiaoke Shang, **Long Ma**<sup>†</sup>, Weimin Wang. *Enhancing Underwater Images via Resonant Fusion*. **IEEE TIP** 2026. (JCR-Q1, IF: 15.3)
- Weimin Wang, Ruifeng Nie, Yingchi Liu, **Long Ma**<sup>†</sup>, Chengpei Xu, Qi Jia, Yu Liu, Na Lei. *SWG-Fusion: Soft Weather-guided Multimodal Fusion with VLM-assistance for BEV Object Detection under Harsh Weather*. **Pattern Recognition** 2026, 178: 113398. (JCR-Q1, IF: 9.1)
- Xinwei Xue, Jincheng Yuan, Tengyu Ma, **Long Ma**<sup>†</sup>, Yi Wang. *Degradation-Decoupled and Semantic-Aggregated Cross-space Fusion for Underwater Image Enhancement*. **Information Fusion** 2025, 118: 102927. (JCR-Q1, IF: 17.4)
- Xinwei Xue, Haoze Zheng, Yuechao Gao, Tengyu Ma, **Long Ma**<sup>†</sup>, Qi Jia. *Crossing the Chasm: A Practical Architecture Augmentation for*

*Low-Quality Object Detection*. **Neurocomputing** 2025, 625: 129553. (JCR-Q1, IF: 6.7)

- Chengpei Xu, Hao Fu, **Long Ma**<sup>†</sup>, Wenjing Jia, Chengqi Zhang, Feng Xia, Xiaoyu Ai, Binghao Li, Wenjie Zhang. *Seeing Text in the Dark: Algorithm and Benchmark*. **ACM Multimedia** 2024: 2870-2878.
- Xinwei Xue, Jia He, **Long Ma**<sup>†</sup>, Yi Wang, Xin Fan, Risheng Liu. *Best of Both Worlds: See and Understand Clearly in the Dark*. **ACM Multimedia** 2022: 2154-2162.
- Yuxin Feng, **Long Ma**, Xiaozhe Meng, Fan Zhou, Risheng Liu, Zhuo Su. *Advancing Real-World Image Dehazing: Perspective, Modules, and Training*. **IEEE TPAMI** 2024, 46(12): 9303-9320. (JCR-Q1, IF: 20.4)
- Risheng Liu, **Long Ma**, Tengyu Ma, Xin Fan, Zhongxuan Luo. *Learning with Nested Scene Modeling and Cooperative Architecture Search for Low-Light Vision*. **IEEE TPAMI** 2022, 45(5): 5953-5969. (JCR-Q1, IF: 20.4)
- Risheng Liu, **Long Ma**, Xiaoming Yuan, Shangzhi Zeng, Jin Zhang. *Task-Oriented Convex Bilevel Optimization with Latent Feasibility*. **IEEE TIP** 2021, 31: 1190-1203. (JCR-Q1, IF: 15.3)
- Risheng Liu, **Long Ma**, Yiyang Wang, Lei Zhang. *Learning Converged Propagations with Deep Prior Ensemble for Image Enhancement*. **IEEE TIP** 2020, 28(3): 1528-1543. (JCR-Q1, IF: 15.3)
- Nan An, **Long Ma**, Guangchao Han, Xin Fan, Risheng Liu. *Striving for Faster and Better: A One-Layer Architecture with Auto Re-parameterization for Low-Light Image Enhancement*. **IEEE TCSVT** 2025, 35(8): 7455-7470. (JCR-Q1, IF: 10.8)

## Awards and Honors

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- 2026 **CVPR Compute Transparency Champion Award** (Corresponding Author)
- 2025 **World's Top 2% Scientists** (Stanford University & Elsevier) (1/1)
- 2025 **ICML Outstanding Reviewer** (1/1)
- 2024 China Society of Image and Graphics (CSIG) Doctoral Dissertation Incentive Program Nomination Award (1/1)
- 2023 ACM Dalian Doctoral Dissertation Award (1/1)
- 2023 The Excellent Paper Award by The Chinese Journal of Image and Graphics (1/3)
- 2023 First Prize of Liaoning Natural Science Academic Achievement Award (3/5)
- 2022 PRCV Best Poster Award (2/3)

## Research Projects

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- 2024-2025, **Sole Chief Investigator**, *Multi-Task Joint Learning Methods for Underwater Visual Perception*, Postdoctoral Fellowship Program (Grade B) of China Postdoctoral Science Foundation, CNY 360,000.  
Supported by this grant, I have published **eight papers** in top-tier conferences and journals (e.g., TPAMI, CVPR, ACM Multimedia). Notably, our CVPR 2025 work maintains state-of-the-art visual enhancement performance in foggy, underwater, and other adverse conditions while **reducing model size by more than 50%**, thereby substantially improving its practical deployability.
- 2024-2025, **Sole Chief Investigator**, *Visual Analysis and Processing under Harsh Environments*, Young Tech Star of Dalian Science and Technology Innovation Foundation, CNY 100,000.
- 2024-2025, **Sole Chief Investigator**, *Visual Perception and Learning in Adverse Conditions*, General Program of the China Postdoctoral Science Foundation, CNY 80,000.

## Academic Service

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- **Program Committee Member**: CVPR, ICCV, ICML, NeurIPS, ICLR, IJCAI, AAAI, ACM Multimedia, ECCV, etc.
- **Journal Reviewer**: IEEE TPAMI, IJCV, IEEE TIP, IEEE TNNLS, IEEE TMM, IEEE TCSVT, IEEE TGRS, etc.